

Name

Midterm

1. Briefly explain the three types of niches we discussed (bullet points are fine)

Grinnellian (Abiotic) Niche:

Eltonian (Biotic) Niche:

Hutchinsonian (n-dimensional) Niche:

2. In ecology, the principle of limiting similarity suggests that (circle one):
 - a. Species with identical niches can coexist indefinitely in the same environment.
 - b. Species that are too ecologically similar will compete strongly and one may be driven to extinction
 - c. More similar species always have a competitive advantage over less similar ones.
 - d. Competition does not influence species diversity or community structure.

3. Briefly describe the latitudinal diversity gradient.

4. What was one other diversity gradient we discussed?

10. Which explanations have been proposed to account for why larger areas tend to support more species? (circle all that apply)

- a. They contain a greater diversity of habitats.
- b. Species in larger areas have a lower likelihood of extinction.
- c. Larger areas have a higher rate of colonization.
- d. Larger areas have fewer predators, allowing more species to survive.
- e. The presence of more species is due purely to chance.
- f. Larger areas have a more stable climate, which directly increases species diversity.

11. Which of the following are common explanations for why species diversity is higher in the tropics? (circle all that apply)

- a. The tropics have a more stable climate over time, reducing extinction rates.
- b. Higher primary productivity supports more species and larger populations.
- c. The tropics have experienced fewer glaciations, allowing for longer periods of uninterrupted evolution.
- d. Species in the tropics evolve more slowly due to the warm climate.
- e. Increased competition in the tropics leads to fewer species coexisting.

12. Draw a picture of your favorite species.

13. Closely related species are (more / less) likely to share diseases (circle one)

14. The theory of Island Biogeography is focused on the impacts of island distance and area on which two rates?

_____ and _____

15. Match each of the following ecosystem services to the correct category: Supporting, Provisioning, Regulating, or Cultural.

Ecosystem Services:

- a. Water purification
- b. Recreation and ecotourism
- c. Pollination
- d. Food production
- e. Climate regulation
- f. Soil formation
- g. Spiritual or aesthetic value
- h. Flood control
- i. Nutrient cycling
- j. Raw materials (e.g., timber, fiber)

Categories:

Supporting: _____

Provisioning: _____

Regulating: _____

Cultural: _____

16. There are two ways in which diversity may drive increases in productivity. Briefly explain each of them:

Sampling Effect:

Niche Complementarity:

17. Diversity is thought to generally (increase / decrease) productivity? (Circle one)

18. Diversity is thought to generally (increase / decrease) nutrient uptake rates? (Circle one)

19. Diversity is thought to generally (increase / decrease) ecosystem stability? (Circle one)

20. Diversity is thought to generally (increase / decrease) invasibility? (Circle one)

21. How is it that you can have constant (or increasing) species richness despite species extirpations from a community?

22. Do biodiversity experiments, theory, and observational studies agree?

- a. Yes
- b. No
- c. Sometimes

23. According to the study we discussed in class, is biodiversity *per se* usually the most important factor in driving ecosystem function in natural systems?

- a. Yes
- b. No

24. Why is the Exponential/Malthusian model of population growth ($N_t = N_0 e^{rt}$) unrealistic in most situations?

25. Which of these factors could cause population growth rates to decline? (circle all that apply)

- a. Competition
- b. Aggression
- c. Disease
- d. Predation
- e. Accumulation of waste

26. Which of these might cause an Allee effect (a.k.a. positive density dependence)? (circle all that apply)

- a. High population growth rate
- b. Shared predator defense
- c. Niche partitioning
- d. Cooperative foraging
- e. Environmental modification

27. What does carrying capacity tell us about a population? (circle one)

- a. Total biomass
- b. Maximum number of individuals
- c. Number of species
- d. Predator density
- e. Mutualist benefit

28. Draw the general shape of a **Type I** and a **Type II** functional response curve. The x-axis should be prey density, the y-axis should be feeding rate. Make sure to label the curves.

29. The following equation describes energy intake rate (E/T) in terms of λ_i (prey species i encountered per unit time), E_i (energy gain from prey species i), p_i (the probability of pursuit and capture of species i) and h_i (handling time of prey species i). Using this equation, explain **two** ways in which a species can increase its energy intake rate.

$$\frac{E}{T} = \frac{\sum_i^k \lambda_i E_i p_i}{1 + \sum_i^k \lambda_i h_i p_i}$$

30. Fill in the blanks (options: >, <, =). In order to coexist:

- i. α_{ii} ____ α_{ji}
- ii. α_{jj} ____ α_{ij}

31. Explain what the above mathematical relations mean in biological terms.

32. In competition for a single resource, the species that requires the (most / least) of that resource will win. (Circle one)

33. In apparent competition with a single predator, the species that can tolerate the (most / least) predation will win. (Circle one)

34. Coexistence requires (circle one):

- a. a tradeoff
- b. Interspecific competition to exceed intraspecific competition
- c. Intraspecific competition to exceed interspecific competition
- d. Two of the above
- e. None of the above
- f. All of the above

35. Briefly describe one of the tradeoffs we discussed in regard to coexistence (or draw a figure if you prefer).

36. In the Sonoran desert, there are winter annual plants that preferentially germinate when a certain amount of rain is received, otherwise remaining dormant for years (decades, in some cases). What type of coexistence mechanism is most likely? (circle one)

- a. Resource partitioning
- b. Relative nonlinearity of competition
- c. Predator partitioning
- d. Storage effect

37. Explain how one of the concepts we discussed in class could be used in an applied context (e.g., conservation biology, agriculture, disease dynamics, etc.). (4 pts)